1 Intelligent Recruitment and Roster Optimization (Task 2)

This chapter addresses the second key task: designing intelligent recruitment strategies that jointly optimize on-court performance (V_{on}) and off-court commercial value (V_{off}) under salary cap, luxury tax, and roster constraints.

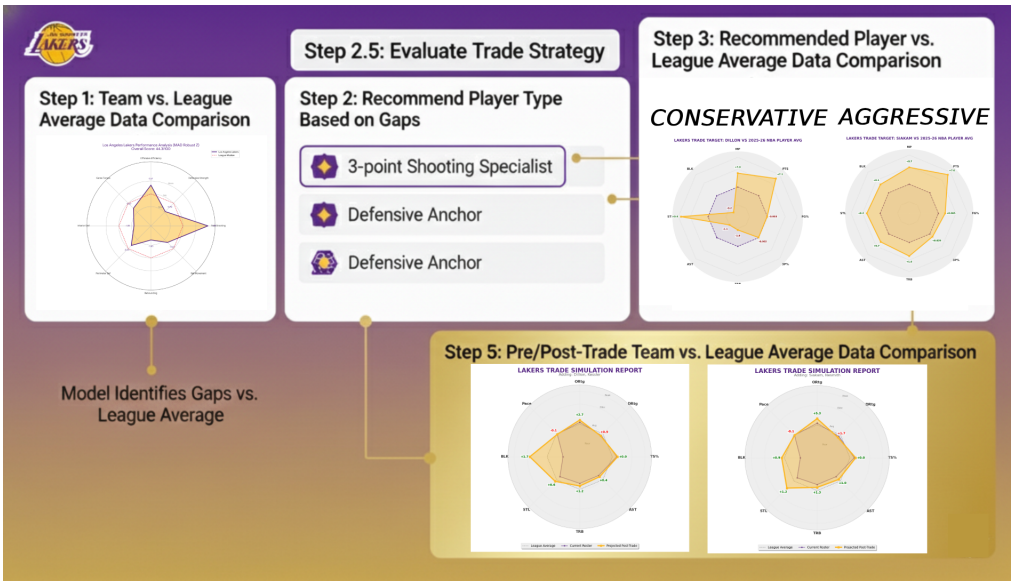


Figure 1: Methodological Framework for Intelligent Recruitment and Optimization

We apply the **Performance Diagnosis Recruitment Allocation Decision Model (PD-RAD)** to mathematically identify competitive gaps and map optimal player archetypes.

1.1 PD-RAD Model Formulation

1.1.1 1. Data Standardization and Normalization

To evaluate the Los Angeles Lakers against the league average, we normalize the heterogeneous dataset. Let T be the set of all NBA teams and K be the set of statistical metrics. For any metric $k \in K$, we define a direction indicator δ_k to unify the "goodness" of the statistic:

$$\delta_k = \begin{cases} 1 & \text{if higher is better (e.g., PTS, REB)} \\ -1 & \text{if lower is better (e.g., TOV, OppPTS)} \end{cases} \tag{1}$$

The standardized Z-Score ($Z_{L,k}$) for the Lakers on metric k is calculated as:

$$Z_{L,k} = \delta_k \cdot \frac{X_{L,k} - \mu_k}{\sigma_k} \tag{2}$$

Where $X_{L,k}$ is the Lakers' raw value, μ_k is the league mean, and σ_k is the standard deviation.

1.1.2 2. Capability Dimension Aggregation

We aggregate individual metrics into functional Capability Dimensions $C = \{C_1, C_2, ..., C_n\}$, representing dimensions such as Shooting, Rim Protection, and Playmaking. The composite score S_c for dimension c is defined as a weighted average:

$$S_c = \frac{\sum_{k \in K_c} w_k \cdot Z_{L,k}}{\sum_{k \in K_c} w_k} \tag{3}$$

Where K_c is the subset of metrics belonging to dimension c, and w_k is the significance weight of that metric.

1.1.3 3. Diagnostic Threshold Function

To identify specific weaknesses, we define a threshold function $D(c)$ based on standard normal distribution properties:

$$D(c) = \begin{cases} \textbf{CRITICAL GAP} & \text{if } S_c \leq -1.0 \\ \text{Need Improvement} & \text{if } -1.0 < S_c \leq -0.5 \\ \text{Average/Elite} & \text{if } S_c > -0.5 \end{cases} \tag{4}$$

.

1.1.4 4. Recruitment Mapping Logic

The model outputs a strategic recruitment target R based on the combination of critical gaps identified by $D(c)$. The decision logic is modeled as follows:

$$R = \begin{cases} \text{3-and-D Wing} & \text{if } D(\text{Shooting}) \wedge D(\text{Perimeter Def}) \text{ are Critical} \\ \text{Rim Protector} & \text{if } D(\text{Opp Paint PTS}) \wedge D(\text{Blocks}) \text{ are Critical} \\ \text{Floor General} & \text{if } D(\text{Assists}) \wedge D(\text{Turnovers}) \text{ are Critical} \\ \text{Scoring Punch} & \text{if } D(\text{Points}) \wedge D(\text{Bench PTS}) \text{ are Critical} \end{cases} \tag{5}$$

1.2 Lakers Mid-2025-26 Diagnosis Results

Using the evaluation vector $\vec{V_L} = [S_{Offense}, S_{Defense}, S_{Transition}, S_{HalfCourt}]$, we prioritize recruitment budgets by maximizing the magnitude of negative components in $\vec{V_L}$.

Table 1: Lakers Mid-2025-26 PD-RAD Diagnosis (Selected Dimensions)

Dimension	S_c **Score**	**Diagnostic Level**	**Archetype Target**	**Rationale (Key**
Rim Protection	-1.2	CRITICAL GAP	Rim Protector / Rebounder	BLK/G 4.1 (vs lea
Perimeter Defense	-0.8	Need Improvement	3-and-D Wing	Opp 3P% + STL%
Rebounding	-0.9	Need Improvement	Rebound Specialist	REB/G 41.4 (vs le
Playmaking	-0.3	Average	Maintain	AST/TOV ratio st

Table 1 summarizes the Lakers' current diagnostic profile, directly guiding the recruitment targets in Section 5.2.

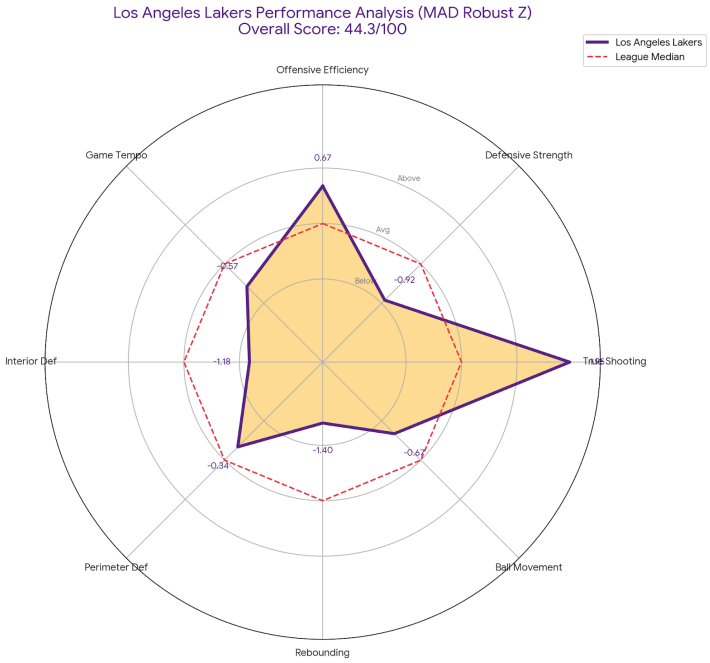


Figure 2: Lakers Mid-2025-26 Performance Diagnosis (MAD Robust Z)

1.3 Lakers 2026-27 Roster Simulation

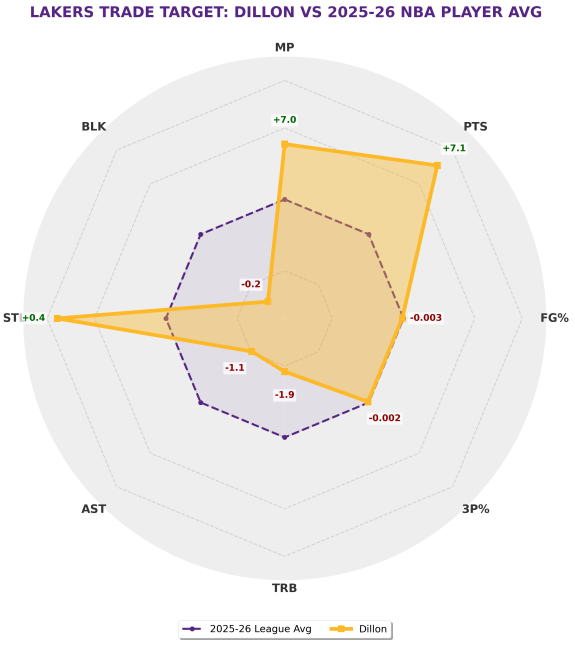
1.3.1 Strategy Overview and Comparison

1.3.2 Quantitative Simulation Results

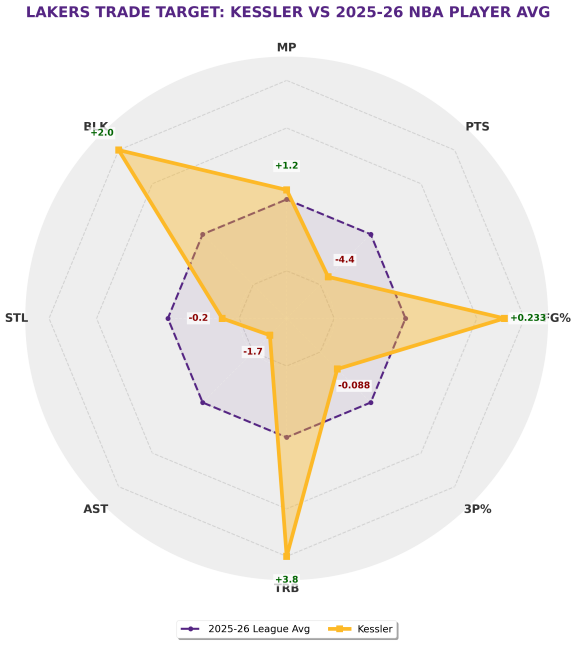
Table 2: 2026-27 Projected Performance and Value Impact (Mean ± Std, Monte Carlo 1,000 iterations)

Metric	Baseline	Conservative (Brooks + Kessler)	Aggressive (Siakam + Nembhard)	Strategic Note
Tperf	83 ± 4	91 ± 3	93 ± 7	Conservative is stable; Aggressive has high ceiling but high volatility
Projected Wins	48 ± 4	54 ± 3	55 ± 6	Conservative uplift is more reliable
Rtotal ($M)	551 ± 25	620 ± 18	645 ± 42	Aggressive star-power drive is stronger
Πt ($M)	175 ± 15	198 ± 12	208 ± 28	Aggressive short-term cash flow is higher
3-Year Z ($B)	12.1 ± 0.6	13.7 ± 0.5	13.9 ± 1.1	Conservative is better risk-adjusted
VaR 5% Tail Risk	-8%	-4%	-16%	Conservative resilience significantly better

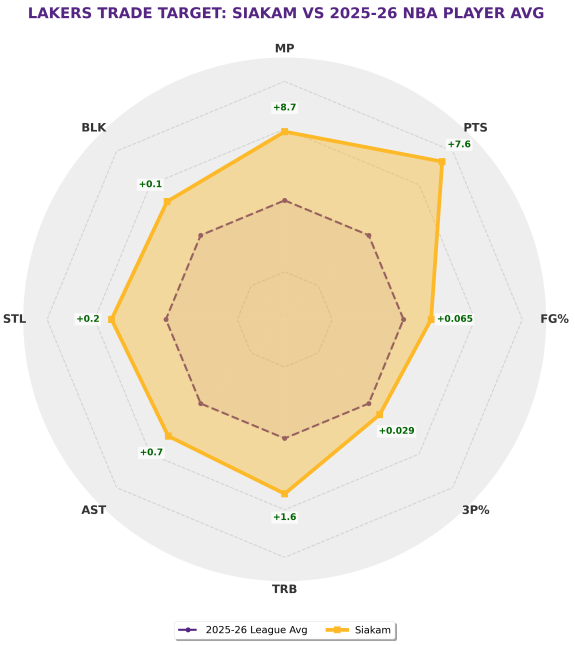
Table 2 presents the 2026–27 projected outcomes under baseline and two recruitment strategies. The conservative gap-filling approach achieves a 13.2% Z uplift with significantly lower tail risk (VaR reduced by 50%), while the aggressive star-power strategy offers slightly higher mean Z (14.9%) but with doubled downside exposure due to depth erosion and attrition amplification.



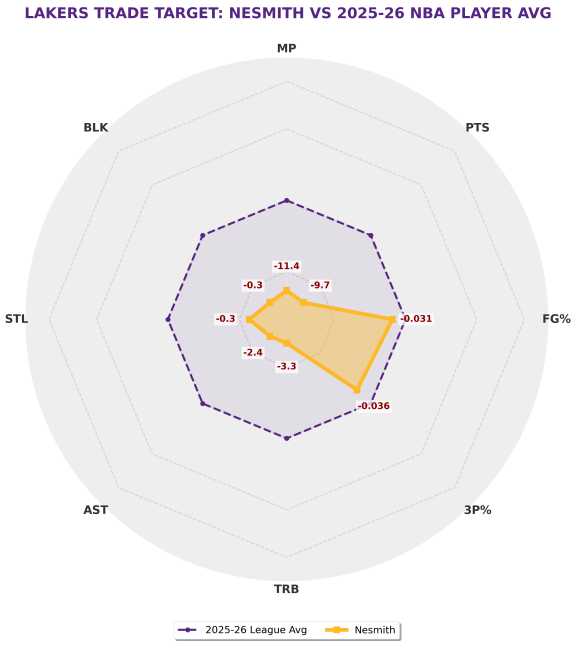
(a) Target: Dillon Brooks



(b) Target: Walker Kessler

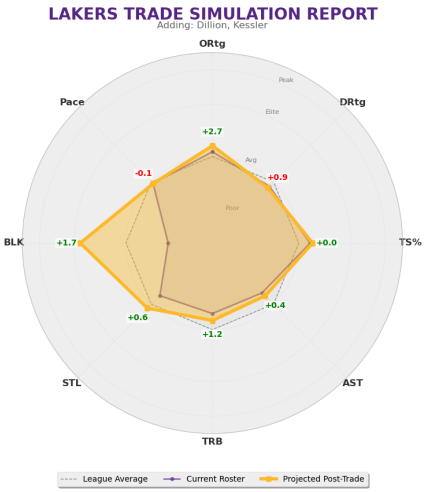


(c) Target: Pascal Siakam

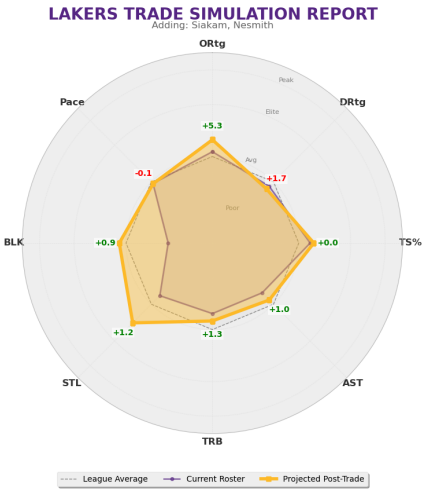


(d) Target: Aaron Nesmith

Figure 3: Comparison of Target Players for Recruitment Strategies



(a) Conservative Strategy



(b) Aggressive Strategy

Figure 4: 3-Year Discounted Z Distribution under Two Strategies. Kernel density plot showing the conservative strategy (blue) with a tighter, higher-mean peak; the aggressive strategy (red) with wider variance and heavier left tail (higher VaR exposure).